MILITARY SPECIFICATION
DELAY ELEMENT, CARTRIDGE, M13 ASSEMBLY

1. SCOPE

1.1 This specification covers Delay Element, Cartridge, M13 Assembly, for use in cartridges for propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-B-550	- Barium Chromate
MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing Aeronautical and Associated Equipment General Specification for
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys
MIL-C-13739	- Composition Delay

Picatinny Arsenal

PA-PD-451	- Boron

STANDARDS

Military

MIL-STD-105	- Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card

FSC 1377

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DRAWINGS

Munitions Command

D8596113	- Delay Element, Cartridge, M13 Assembly
B8826728	- Ammunition, Container, Metal, M21 for Delay Elements M5 or M8
B8826729	- Box, Packing, Ammunition for Delay Elements M5 or M8 in Metal Container M21.
IEL-8596113	- Inspection Equipment List for Delay Element, Cartridge, M13, Assembly

(Copies of specifications standards, drawings and publications required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.4 Vibration.- The delay element shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer or delay charge or derangement of any components.

3.5 Functional.

3.5.1 The delay element shall not fail to fire.

3.5.2 Delay charge.- The delay charge shall consist of a composition Borium Chromate as specified in MIL-B-550 and Boron as specified in PA-PD-451, or as specified in MIL-C-13739, in the proportion and quantity fixed by the contractor to produce the specified delay time requirements prescribed in Table I (see 6.2).

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Table I

Temperature	Delay Time
-65° to -70°F	The mean value plus 3 times the standard deviation shall not exceed 4.2 seconds.
160° to 165°F	The mean value minus 3 times the standard deviation shall be not less than 2.3 seconds.

3.6 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawing, specifications and the following:

3.6.1 Metal defects.- The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign material.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 23 delay bodies, 23 primer retainers and 30 delay element assemblies

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4.2.1 (Cont'd)
to be tested as indicated in Table II and 10 sets of unassembled components (including 10 primers) for dimensional and visual examination.

Table II

Paragraphs		Test	Sample Size	Type of Test	
Requirement	Test			Dest	Non-Dest
3.3	4.4.1	Protective finish	23 delay bodies	X	-
			23 Primer retainers	X	-
3.4	4.4.2	Vibration	12 ea. from Ballistics sample	-	X
3.5	4.4.3	Ballistics	30	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistics test shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

Table III

Temperature	Not Vibrated	Vibrated
-65° to -70°F	9	6
160° to 165°F	9	6

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

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4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency an ammunition lot shall consist of 1200 delay element assemblies maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process.
b. Delay charge from not more than one lot.
c. Primers from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defect within a major or minor class, the criteria in Table V shall be used. All non-conforming material shall be rejected. Four power magnification (4X) may be used to perform visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

Table IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table V

Sample Size for Major Defects	Total allowable defect per Sample Size			
	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over - 43	6	8	11	14

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Table V (Cont'd)

Sample Size for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 43	6	8	11	14
over - 43	8	11	17	20

4.3.2.1 Primer, M42, Dwg C8799925 covering a detail of Dwg D8596113.

Defect	Method of Inspection
CRITICAL	
1. Mixed primers	Visual
2. Primer pellet broken, cracked or dislodged	Visual
3. Primer foil discolored	Visual
4. Metal defective	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.2 Body, Delay, Dwg C8594081, covering a detail of Dwg D8596113.

CRITICAL	
1. Diameter of flash hole	Gage
2. Metal defective	Visual
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of body with pitch diameter of thread	Gage

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
104. Diameter of thread relief	Gage
105. Diameter of step at thread relief, min.	Gage
106. Diameter of cavity, max.	Gage
107. Diameter of body, min.	Gage
108. Depth of cavity, max.	Gage
109. Length from cavity face to body diameter	Gage
110. Thread damaged or not full	Visual
111. Protective finish inadequate or defective	Visual
112. Burr	Visual
113. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of cavity, min.	Gage
203. Diameter of body, max.	Gage
204. Diameter of step at thread relief, max.	Gage
205. Diameter of wrench holes	Gage
206. Depth of cavity, min.	Gage
207. Depth of wrench holes	Gage
208. Total length	Gage
209. Length of thread pilot	Gage
210. Length of step at thread relief	Gage
211. Length of thread relief	Gage
212. Location of wrench holes	Gage
213. Radius or chamfer missing or incorrect	Visual
214. Surface finish improper	Visual

4.3.2.3 Gasket, Aluminum, Dwg B8594082 covering a detail of Dwg D8596113.

CRITICAL	
None defined	
MAJOR	
101. Diameter, inside, max.	Gage
102. Metal defective	Visual
103. Burr	Visual
104. Foreign matter	Visual

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter, inside, min.	Gage
202. Diameter, outside	Gage
203. Total thickness	Gage
204. Surface finish, improper	Visual

4.3.2.4 Primer Retainer, Dwg C8594083, covering a detail of Dwg D8596113.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of internal thread, max.	Gage
102. Minor diameter of internal thread, max.	Gage
103. Concentricity of head with body	Gage
104. Concentricity of primer pocket with body	Gage
105. Concentricity of pitch diameter of thread with body	Gage
106. Concentricity of recess around primer pocket with primer pocket	Gage
107. Diameter of primer pocket flash hole	Gage
108. Diameter of body, min.	Gage
109. Diameter of head	Gage
110. Diameter of primer pocket	Gage
111. Diameter at base of angled recess	Gage
112. Diameter of recess around primer pocket	Gage
113. Depth of primer pocket	Gage
114. Depth of cavity	Gage
115. Depth of recess around primer pocket	Gage
116. Depth of chamfer at primer pocket	Gage
117. Length of effective thread, min.	Gage
118. Location of datum diameter	Gage
119. Thread damaged or not full	Visual
120. Metal defective	Visual
121. Protective finish inadequate or defective	Visual
122. Surface finish of primer pocket improper	Visual
123. Burr	Visual
124. Foreign matter	Visual

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of internal thread, min.	Gage
202. Diameter of body, max.	Gage
203. Total length	Gage
204. Length of wrench flats	Gage
205. Width across wrench flats	Gage
206. Location of wrench flats	Gage
207. Radius or chamfer missing or incorrect	Visual
208. Surface finish, improper, except as otherwise classified	Visual

4.3.2.5 Delay Element, M13, Assembly, Drawing D8596113.

CRITICAL	
1. Any part missing	X-ray and Visual 1/
2. Improper assembly	X-ray and Visual 1/
3. Primer not properly seated	X-ray and Visual 1/
4. Mixed ammunition	Visual
5. Flash hole missing or obstructed	Visual
6. Primer uncrimped or seal missing	Visual
7. Marking missing, incorrect or illegible	Visual
8. Delay charge not visible through flash hole	Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Diameter of body, max.	Gage
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Total length, max.	Gage
202. Lacquer missing or inadequate	Visual

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1/ Prior to assembly into a cartridge the delay element shall be x-rayed with the sides of the element perpendicular to the rays. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.6 Ammunition, Container, Metal, M21 for Delay Element M5 or M8, Dwg B8826728.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Sealing inadequate, incomplete or incorrect	Visual
205. Container cut or damaged	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.7 Box, Packing, Ammunition for Delay Element, M5 or M8 in metal container M21, Dwg B8826729.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual

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4.3.2.7 (Cont'd)

MINOR (Cont'd)

Defect	Method of Inspection
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Contents incorrect	Visual
207. Marking missing, incorrect or illegible	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.- Twenty three units of each anodized component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.3.3.2 Vibration.- Twelve of the 30 assembled x-rayed delay elements of the initial production sample shall be subjected to vibration in accordance with 4.4.2. After vibration the 12 vibrated delay elements shall be x-rayed as directed in 1/ of 4.3.2.5. Failure of any unit to comply with the requirements of 3.4 shall be cause for rejection of the lot. When there is a change in the method of charging the delay element or in the lot of primers, twelve additional delay elements shall be selected and tested as indicated herein.

4.3.3.3 Functional.- The following tests shall be performed by the Government.

4.3.3.3.1 Ballistics.- The 30 test delay elements from each production lot shall be grouped and temperature conditioned in accordance with 4.4.3.1.1 and Table IV. Failure of any cartridge to comply with the requirements of 3.5 shall be cause for rejection of the lot.

Table VI

Temperature	Sample Size
-65° to -70°F	15
160° to 165°F	15

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 8596113.

4.4 Test methods and procedures.

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4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539.

4.4.2 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Functional.

4.4.3.1 Ballistics.- Delay elements of the sample shall be grouped and temperature conditioned in accordance with 4.4.3.1.1 and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded the sample unit shall be reconditioned in accordance with 4.4.3.1.1.

4.4.3.1.1 Temperature conditioning.- A standard inert delay element, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy delay element to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8826728 and C8826729, except that the marking shall refer to Delay Element, Cartridge, M13 Assembly.

5.2 Data Cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Delay time is defined as the time lapse between the impact of the firing pin on the primer and evidence of gas flow from the vent hole of the delay element.

6.2 Advisory delay charge composition (formulation).- Approximately 96.2 percent Barium Chromate as specified in MIL-B-550 and 3.8 percent Boron as specified in PA-PD-451, with an individual delay element charge weight of approximately 22 grains supplemented with an igniter charge on top of the delay column composed of 90 percent barium chromate and 10 percent boron, weighing approximately 1 grain.

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6.2.1 Alternate advisory charge:- five grains type II, plus 6 grains, type III, plus 5 grains, type II as specified in MIL-C-13739.

Custodian:

Army - MU

Preparing activity:

Army - MU

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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